GIT BRANCH

Definition

A Git branch is a separate line of development and workspace in a repository used to work on changes independently without affecting the main code.

Uses of Git Branch

- Create separate workspace for development
- Develop new features separately
- Fix bugs safely
- Test code without affecting main branch
- Allow multiple developers to work simultaneously
- Merge changes after successful testing

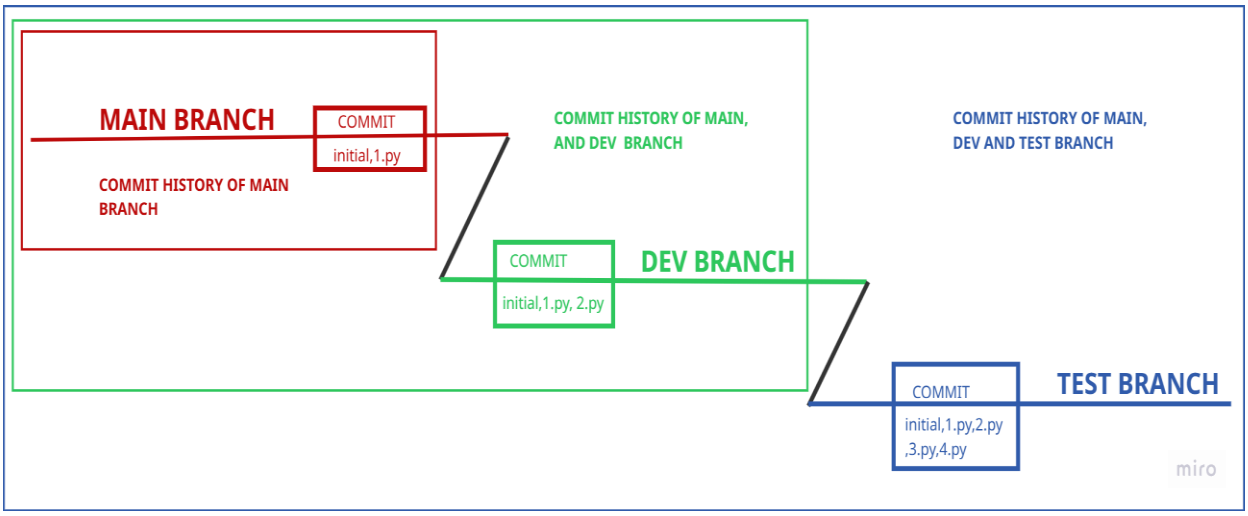
Features of Git Branch

- Lightweight and fast to create
- Multiple branches in one repository
- Independent development environment
- Easy switching between branches
- Supports merging changes into main branch
- Helps in organized and parallel development

Git Branch Commands with Definition

Command	Definition
git branch	Show all branches in the repository
git branch branch-name	Create a new branch
git checkout branch-name	Switch to another branch
git checkout -b branch-name	Create and switch to a new branch
git merge branch-name	Merge a branch into the current branch
git branch -d branch-name	Delete a branch

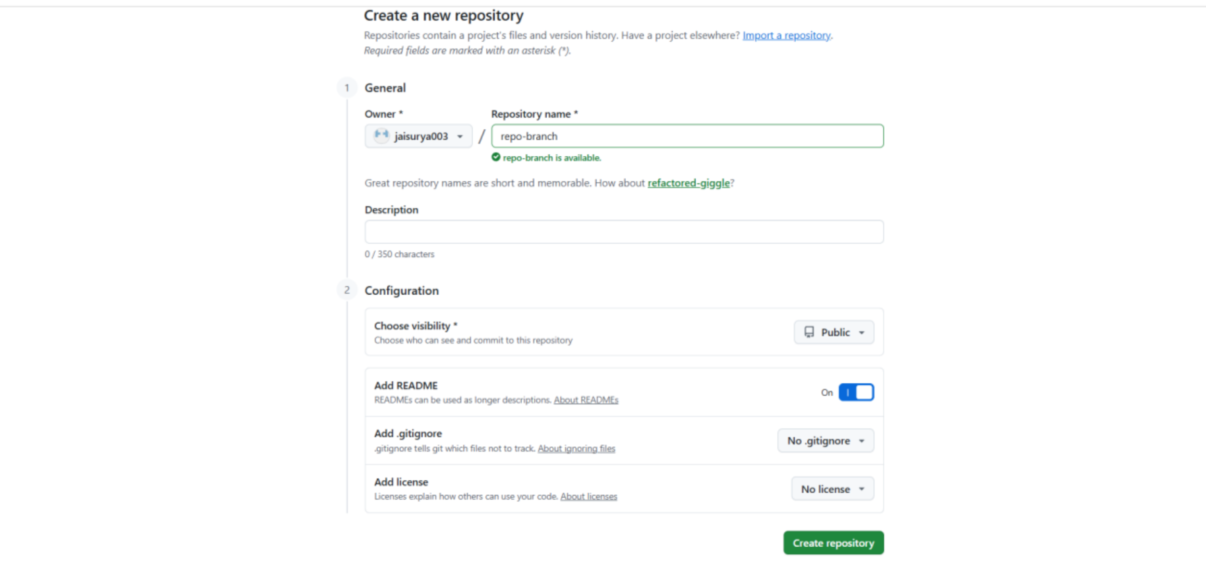
GIT BRANCH ARCHITECTURE



HANDS ON FOR GIT BRANCH

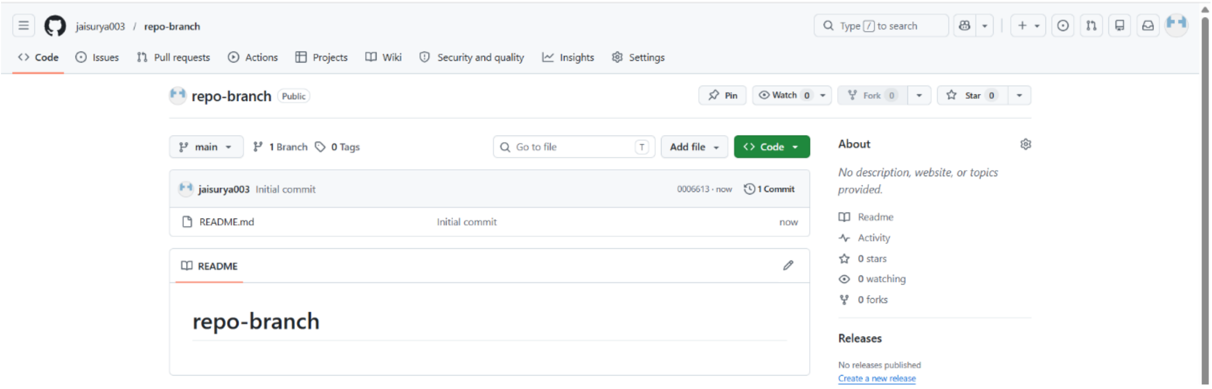
Screenshot: Create New Repository

- Click New repository
- Enter repository name
- Enable Add README file
- Click Create repository



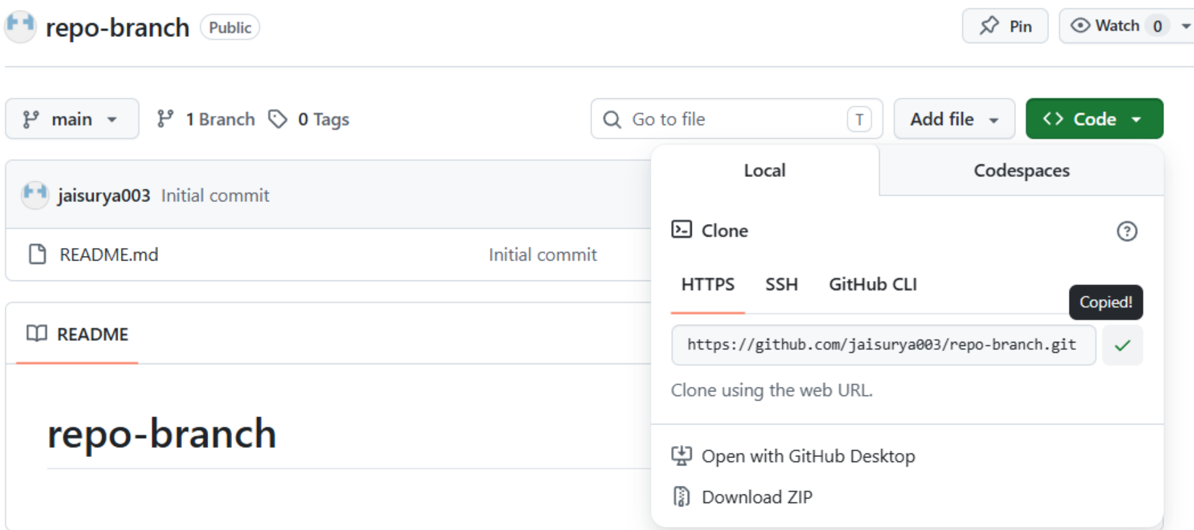
Screenshot: Repository Created

- Repository created successfully
- Repository dashboard opened



Screenshot: Copy Repository Link

- Click Code
- Copy HTTPS repository link



Screenshot: Clone Repository in Git Bash

- Open Git Bash
- Execute cd Downloads
- Execute git clone <repository-link>
- Execute cd <repository-name>

```
MINGW64:/c/Users/JAISURYA/downloads/newfolder/repo-branch

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~
$ cd downloads

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads
$ mkdir newfolder

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads
$ cd newfolder

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder
$ git clone https://github.com/jaisurya003/repo-branch.git
Cloning into 'repo-branch'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder
$ cd repo-branch
```

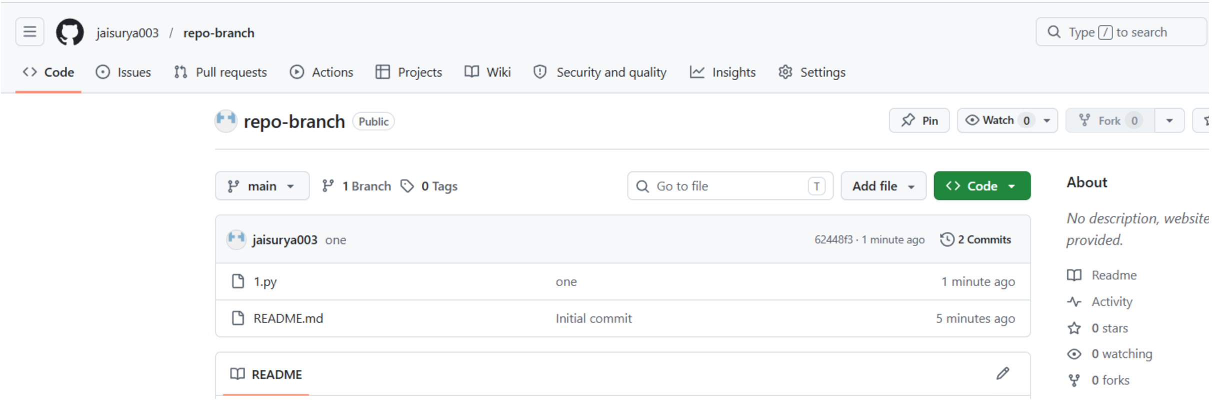
Screenshot: Create and Push File

- Execute touch 1.py
- Execute git add 1.py
- Execute git commit -m "one"
- Execute git push origin main

```
JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder/repo-branch (main)
$ touch 1.py

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder/repo-branch (main)
$ git add 1.py

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder/repo-branch (main)
$ git commit -m "one"
[main 62448f3] one
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 1.py

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder/repo-branch (main)
$ git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 266 bytes | 133.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/jaisurya003/repo-branch.git
   0006613..62448f3  main -> main
```

Screenshot: File Added in Main Branch

- Refresh GitHub repository
- Verify 1.py file is available in main branch



Screenshot: Create Git Branch

- Execute git branch
- Execute git branch dev

```
JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder/repo-branch (main)
$ git branch
* main

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder/repo-branch (main)
$ git branch dev
```

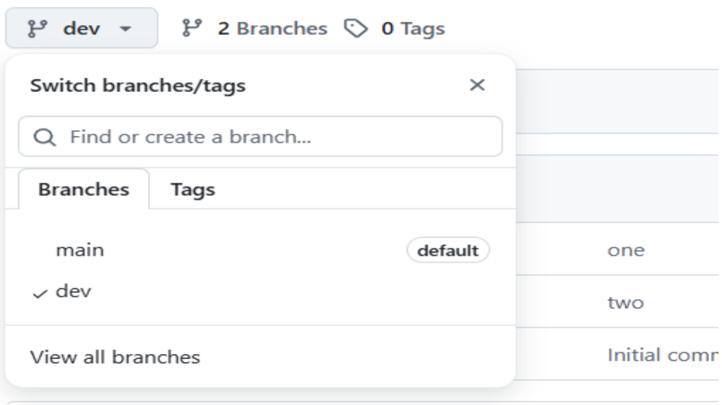
Screenshot: Push Changes to Dev Branch

- Execute git branch
- Execute git checkout dev
- Execute touch 2.py
- Execute git add 2.py
- Execute git commit -m "two"
- Execute git push origin dev

```
JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder/repo-branch (main)
$ git branch
  dev
* main

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder/repo-branch (main)
$ git checkout dev
Switched to branch 'dev'

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder/repo-branch (dev)
$ touch 2.py

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder/repo-branch (dev)
$ git add 2.py

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder/repo-branch (dev)
$ git commit -m "two"
[dev 71214bd] two
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 2.py

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder/repo-branch (dev)
$ git push origin dev
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 261 bytes | 261.00 KiB/s, done.
Total 2 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'dev' on GitHub by visiting:
remote:      https://github.com/jaisurya003/repo-branch/pull/new/dev
remote:
To https://github.com/jaisurya003/repo-branch.git
 * [new branch]      dev -> dev

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder/repo-branch (dev)
$
```

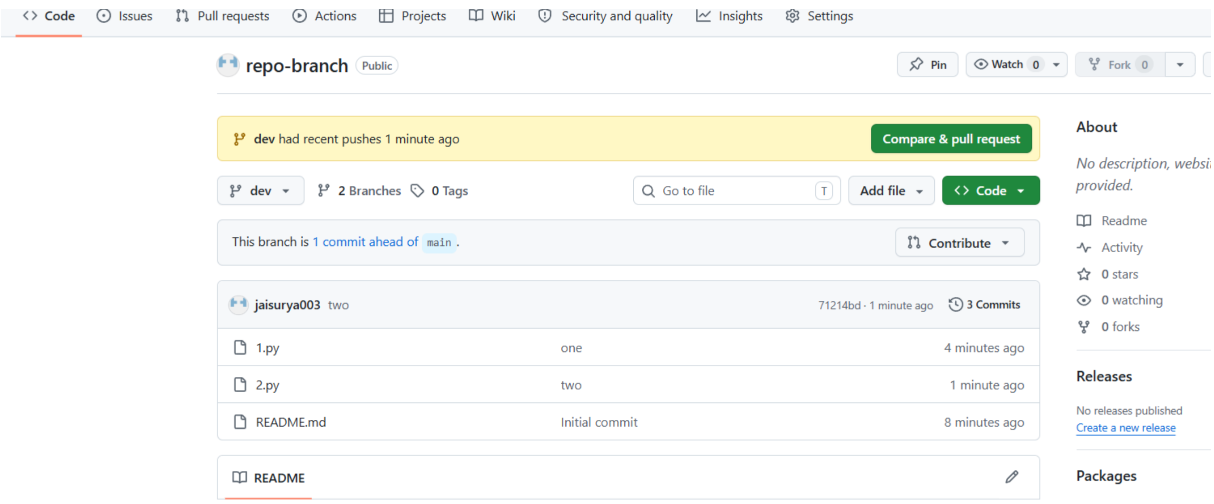
Screenshot: Dev Branch Created in GitHub

- Go to GitHub repository
- Switch branch to dev



Screenshot: Dev Branch Files Verified

- Switch to dev branch on GitHub
- Confirm 1.py, 2.py, and README.md are available in dev branch
- Files successfully present in dev branch



Screenshot: Create and Push Test Branch

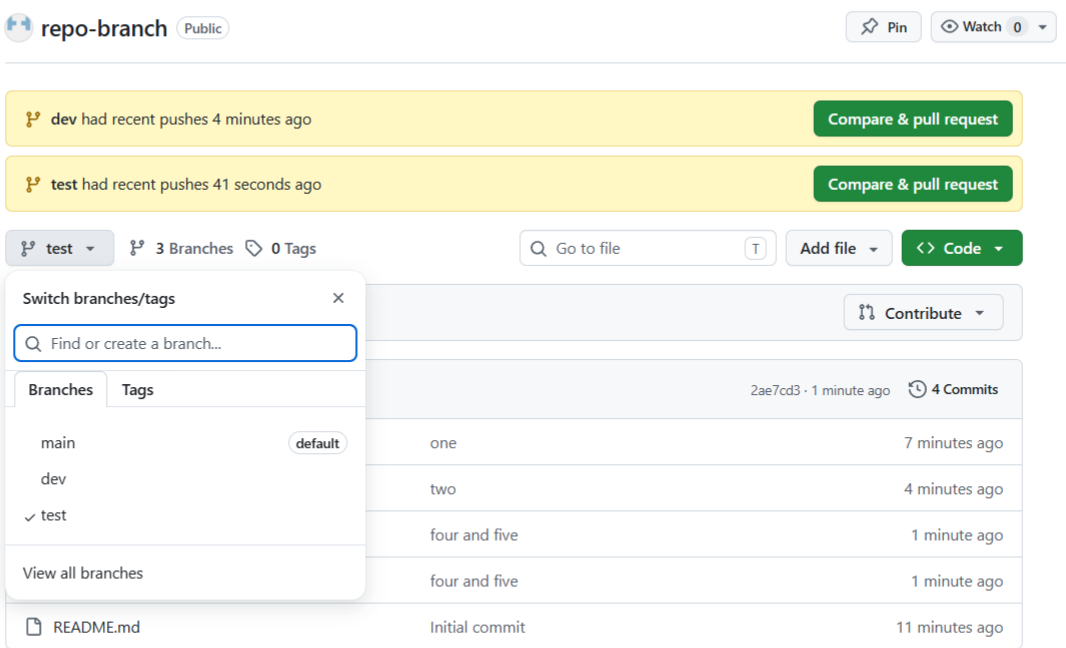
- Execute git checkout -b test
- Create 3.py and 4.py
- Execute git add .
- Execute git commit -m "test branch files"
- Execute git push origin test

```
JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder/repo-branch (dev)
$ git checkout -b test
Switched to a new branch 'test'

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder/repo-branch (test)
$ touch 3.py

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder/repo-branch (test)
$ touch 4.py

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder/repo-branch (test)
$ git add .
```

```
JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder/repo-branch (test)
$ git commit -m "four and five"
[test 2ae7cd3] four and five
 2 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 3.py
 create mode 100644 4.py

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder/repo-branch (test)
$ git push origin test
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 274 bytes | 274.00 KiB/s, done.
Total 2 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'test' on GitHub by visiting:
remote:      https://github.com/jaisurya003/repo-branch/pull/new/test
remote:
To https://github.com/jaisurya003/repo-branch.git
 * [new branch]      test -> test

JAISURYA@LAPTOP-G96D93Q4 MINGW64 ~/downloads/newfolder/repo-branch (test)
```

Screenshot: Test Branch Created in GitHub

- Go to GitHub repository
- Switch to test branch



Screenshot: Test Branch Files Verified

- Switch to test branch on GitHub
- Confirm 1.py, 2.py, 3.py, 4.py, and README.md are available
- All files successfully present in test branch